Jingtao Lei

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Personal Page | Github | Google Scholar

Education

- **Central South University, Dundee International Institute** *Sep 2024 – Jun 2028 (Expected)*
  B.Sc. in Applied Mathematics; Dual-degree programme with the University of Dundee; Cumulative Average: 87.88/100

Research Interests

I work on **world action models**, focusing on how multimodal generative models represent physical states and predict how they evolve under actions. My interests span four questions:

- **Physical scene representations:** Encoding objects, metric distances, motion, and interactions in structured states.
- **Spatiotemporal reasoning:** Understanding how spatial relationships and physical dynamics evolve over time.
- **Action-conditioned prediction:** Learning state transitions and forecasting future states under different actions.
- **Long-horizon consistency:** Updating predictions with new observations to reduce accumulated prediction drift.

Selected Publications

* = corresponding author

[1] **Jingtao Lei***, Hongji Li, Dexiang Shu. (2026). **Layer-Wise Gate-Controlled Prompt Truncation in a Multimodal Chest X-Ray Classifier**. *arXiv preprint* [arXiv] [Code]
Introduces a layer-wise gating mechanism that adaptively truncates visual prompts for multimodal chest X-ray classification.

Research Experience

- **Extending Generative Models toward World Action Models** *Jun 2026 – Present*
  The Chinese University of Hong Kong | *SURP* | *Supervisor: Prof. Shengchao Liu*
  - Built a controlled benchmark of 192 sequences covering single-object motion and two-object interactions to study how generative models represent direction, speed, optical flow, and object contact.
  - Linear probes decoded motion with 1.09° direction error and $R^2 = 0.988$ for speed; complementary 3D future-trajectory prediction experiments reduced mean position error from 2.81 to 1.48 (47%) after LoRA adaptation.
- **Adaptive Visual Prompting for Medical Image Classification** *Jan 2026 – Jul 2026*
  Carnegie Mellon University | *Supervisor: Prof. David P. Woodruff*
  - Developed a layer-wise gating controller for a ViT-based Mixture of Prompt Experts classifier, enabling adaptive visual-prompt truncation [1].
  - Reduced visual-prompt length from six tokens to one (83% fewer tokens), achieving 0.8963 validation accuracy on the IU X-Ray classification task.

Technical Skills

Programming & Scientific Computing: Python, Bash, C++, NumPy, pandas, Matplotlib
Machine Learning: PyTorch, Vision Transformers, LoRA
Research Methods: Controlled benchmark design, representation probing, OOD evaluation, 3D trajectory analysis
Systems & Research Computing: Linux, SSH, Git, Slurm, multi-GPU HPC computing
Independent Study: UC Berkeley CS 61A; MIT 6.031: Software Construction

Honors and Awards

- First Prize, 21st CSU Mathematical Modeling Competition (National Selection) *2026*
- Second Prize, 6th Celian Cup Mathematical Modeling Elite League (Provincial-level Competition) *2026*
- Third Prize, Annual Comprehensive Student Evaluation Award *2024, 2026*
- 97.80/100 (A+), From Combinatorics to Calculus (Prof. Dan Ciubotaru, University of Oxford) *2025*
- AWS Academy Graduate – Cloud Foundations *2025*